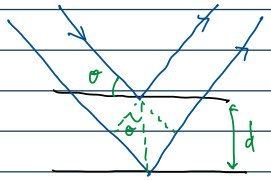
Lecture 6

Thursday, 7 September 2023 10:35 AM

X-ray Diffraction

+ Determine structure of material w/ some long-range order

+ Bragg's law: $\lambda = 2d \sin\theta$



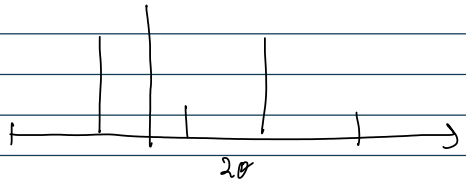
This is fixed → what we need to determine

Specific θ value for a given X-ray source

+ Geometry & intensity will vary on the source (synchrotron vs. lab based)
↓
Quantitative, under reaction conditions

+ Understanding XRD patterns:

$$d_{hkl} = \sqrt{\frac{a^2}{h^2+k^2+l^2}}$$



hkl (100) (110) (111) ....

Intensities of peaks decreases

For simple cubic → certain peaks show up

FCC → 111, 211, 311

BCC → 110, 220

} Based on the group you will have different lines that show up.

Extinction Rule:

Form factor: Scattering by an atom over scattering of an e⁻

↳ This is for each atom & heavier elements will have higher form factor

d spacing is related to the atomic size → so you do know which structure is diffracting in what pattern

We get zero information about the degree of crystallinity

Simulation of XRD patterns → can do that!

→ Rietveld Refinement → when you make a new structure, you use least squares fitting approach

→ Sherrer Equation → Instrument broadening is an issue

$$B(2\theta) = \frac{K\lambda}{L\cos\theta}$$

→ Laue Equations →